SE Lab 3
Architectural Pattern Selection & Component Model

Project Title: Virtual Technical Conference Platform

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Architecture Selection: I chose the Microservices Architecture for the Virtual Technical Conference Platform.

1. Architectural style analysis

Style	Assessment against the Virtual Technical Conference Platform
Layered	This style has clear separation and is easy to build, but presentation, business and data layers ship as one deployable. A keynote pushing 2000 concurrent viewers (NFR-001) would force the scheduling and Q&A code to be replicated with the media code and a fault anywhere in the process takes the whole event down. **Rejected**.
Client–Server	Simple deployment and consistent data, but the single server is both a scalability bottleneck and a single point of failure. A conference runs in a fixed time window, so an outage during a live session cannot be recovered afterwards (NFR-002). **Rejected**.
Microservices	Service-per-capability behind an API gateway. Each capability scales, fails and is deployed independently, which matches the very different load profiles and criticality of scheduling, media delivery and Q&A. **Selected**.

2. Two scenario-specific reasons for Microservices based architecture

Reason 1 - The capabilities have unrelated load curves. NFR-001 requires 720p at under 5 s latency for 2000 attendees on a single keynote, while the Session Manager is serving a handful of Track Chair requests at the same moment. As separate services, only the Live Media & Recording Service is scaled out for the keynote hour while scheduling, registration and Q&A stay at their normal replica count.

Reason 2 - A live event cannot be replayed, so faults must be contained. NFR-002 allows roughly 1.5 minutes of downtime per conference day. Isolating Q&A, registration and media into separate deployables with their own stores means a crash in the Q&A & Moderation Service, or an outage at the external Payment Gateway, degrades one feature while the stream (FR-002) and the join path (FR-003) keep running.

3. Security advantage

Sensitive data is confined to one service and one trust boundary. Card details never enter the platform: the Registration & Payment Service is the only component that holds the PaymentGatewayAPI connection to the external Payment Gateway, and every other service sees only an opaque entitlement token. All external traffic enters through the API Gateway & Identity Service, which validates the token once and injects a scoped identity, so FR-003's rejection of unregistered users is enforced at a single server-side choke point rather than in the client, and the Live Media Service never trusts a client-supplied identity. Internal calls run over mTLS inside the service network, so the blast radius of a compromised service is limited to the interfaces it is authorised to call.

4. Performance benefit

The media path is separated from the request/response path. Live audio and video leave the Live Media & Recording Service straight to the external streaming provider's edge over StreamDelivery, and never traverse the scheduling, registration or Q&A services. A burst of question posts and upvotes on its own WebSocket fan-out (FR-004, 2s visibility) therefore cannot add queueing delay to the video path, which keeps p95 glass-to-glass latency under the 5 s budget of NFR-001. Transcoding and publishing the recording run asynchronously in the same service after the session ends, off the live path.

5. Component diagram

